Credibility of monetary policy

MU and monetary policy credibility

- The choice of independent countries to use recurrent devaluations to adjust for asymmetric permanent shock leads to a loss of control of inflation, which is a primary objective of monetary policy
- Losing price stability then implies losing the credibility of monetary policy
- Indeed, one advantage of joining the EMU has been, for many countries, the recovery of monetary policy credibility

- The relevance of the credibility of monetary policy and of CB reputation

- How to gain monetary policy credibility by joining the EMU

Monetary policy and time inconsistency

- Temporal dynamics that involve some form of temporal (in)coherence in the actions of agents is typical of a wide variety of social and economic interactions.
- Examples
  - Governments issue bonds to be able to spend in excess of revenues and promise to pay pack the debt, but may be tempted to default if a severe crisis strikes
  - A country may enter a fix exchange rate regime but may be tempted to devalue if a severe balance of payment shock strikes
  - To foster innovations a government may announce a zero-tax regime for a particular industry, but then can be tempted to actually collect tax revenues once the industry flourishes
  - To foster studying exams are announced but best solution would then be not to carry them out, to save time and energy
- If announcements lack credibility in the future, they will not be effective→ discourage investors to buy bonds, businesses to engage in R&D and students to study in the first place.

Main references

Kydland e Prescott (1977) «Rules rather than discretion: the inconsistency of optimal plans», Journal of Political Economy

- Nobel prize in 2004 "for their contributions to dynamic macroeconomics: the time consistency of economic policy and the driving forces behind business cycles"

Barro e Gordon (1983) «A positive theory of monetary policy in a naturale rate model», Journal of Political Economy

Barro e Gordon (1983) «Rules, discretion and reputation in a model of monetary policy», Journal of Monetary Economics

Credibility and time inconsistency

Credibility problem

- Government announces a policy move
- But if it is subsequently given an opportunity to obtain a higher (short-run) result, it will attempt to do so, even if this implies “cheating” the public!
- Rational agent know this, anticipate the cheating attempt by the government and will adjust their response functions

Time-inconsistency

- Policy makers find it optimal to announce a commitment to a particular policy and deviate from it after the people have made their decisions based on the announcement.

Why do policy makers benefit by deviating from the announcement?

- Suppose there is a positive short-run relationship between inflation and economic activity (real output or employment), or, which is the same, a negative relationship between inflation and the rate of unemployment (the **Phillips curve**)
- CB can influence inflation and is interested in real objectives (e.g. employment or unemployment), then it may be tempted to exploit this short run relationship in order to achieve a higher real output (or lower unemployment).

Surprise inflation and unemployment in the expectation augmented PC

$\dot{p} = c - bu + \dot{p}^e + \varepsilon$

In the long-run stationary equilibrium

$\dot{p} = \dot{p}^e$ and

$u = \frac{c}{b} = u_N$ Natural rate of unemployment

The short-run Phillips curve can be written as:

$\dot{p} - \dot{p}^e = b\left(\frac{c}{b} - u\right) + \varepsilon$

$\dot{p} - \dot{p}^e = b(u_N - u) + \varepsilon$

Or (ignoring causality issues)

$u = u_N - 1/b(\dot{p} - \dot{p}^e) + \varepsilon$

Note: The natural rate need not be optimal. In fact, the benefits from surprise inflation arise when the policymaker views the natural rate as excessive

Other sources of benefits from surprise inflation: revenue incentive

- Governmental liabilities fixed in nominal terms.
- Real amount of government bonds held by people B_{t-1}/P_{t-1}. These bonds carry the nominal yield, R_{t-1}, which is satisfactory given people's inflationary expectations over the pertinent horizon.
- Surprise inflation depreciates part of the real value of these bonds, which lowers the government's future real expenditures for interest and repayment of principal.
- Quantitatively, this channel is significant for high levels of public debt
- For example, given an outstanding public debt of euro 2000 bil a surprise inflation of 1% would lower the real value of debt by around 20 bil

Agents cannot be fooled for ever

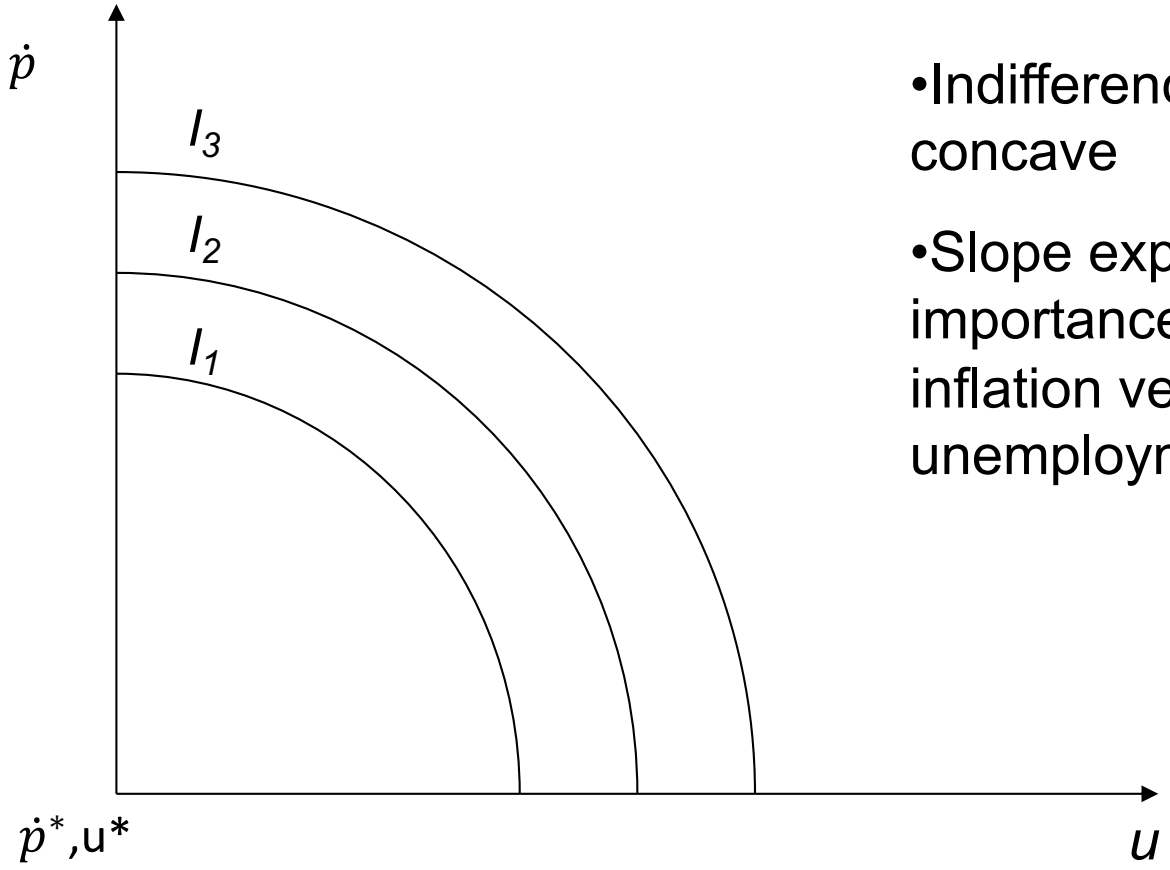
- Economic agents (consumers, firms and other agents in the economy) learn over time and adjust their own inflation expectations.
- Economic agents learn ‘the trick’, i.e. understand that policy makers will try to create surprise inflation to increase real output.
- Once economic agents adjust their expectations, no output (or revenue) gain will be possible even in the short-run.
- Note: Surprise inflation bears benefits but also some costs…
  - Costs of inflation are related to changing prices
  - The model will consider a loss (cost) function to be minimized under a constraint

Preferences of authorities

- Loss function
- Arguments are deviations of inflation and of the rate of unemployment from targets, both deviations are squared.
- Importance attached to fighting inflation vs fighting unemployment may differ since authorities may have different preferences.
- Weights attached to deviations capture the preferences.

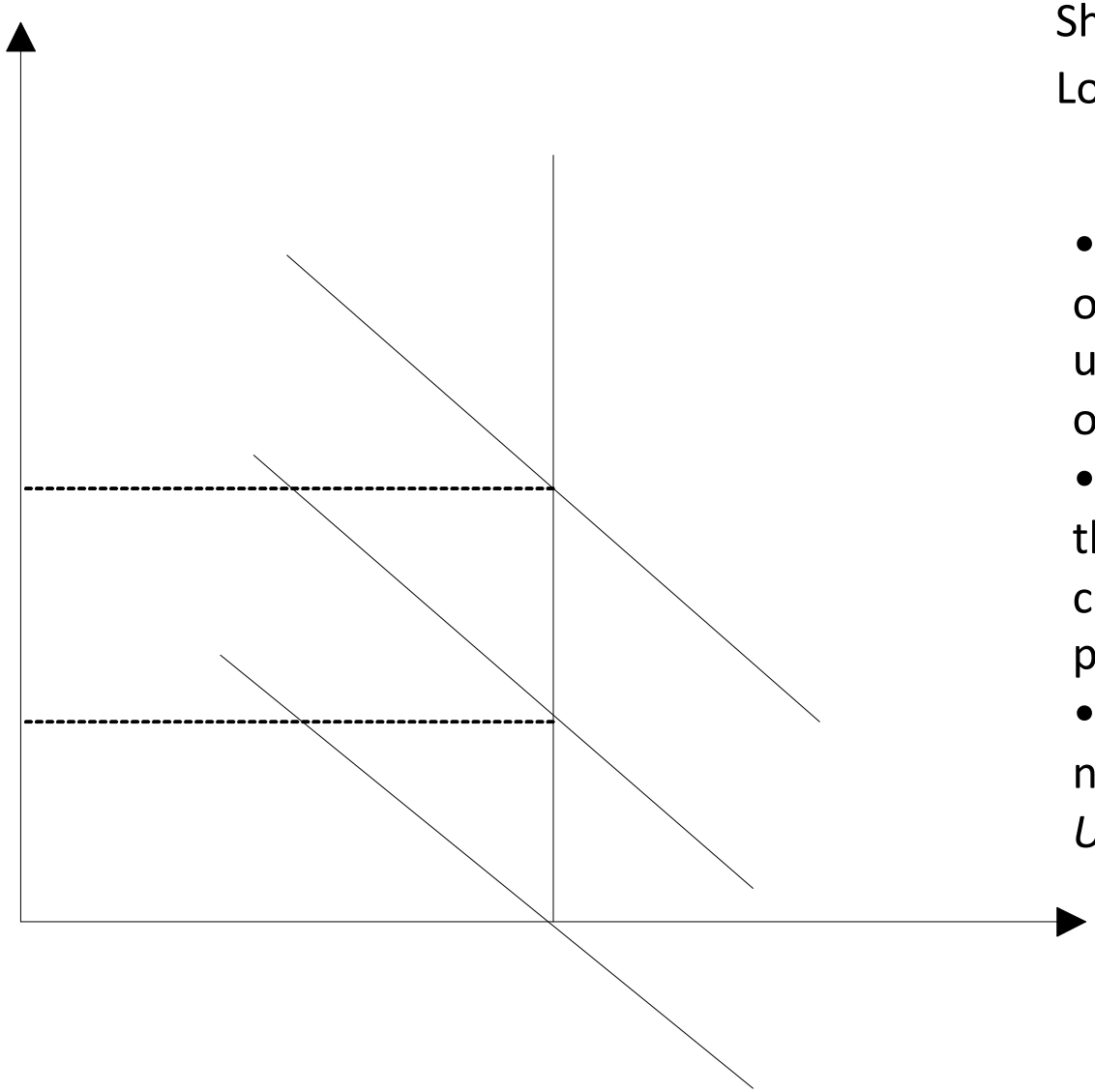
$$L = a_1(\dot{p} - \dot{p}^*)^2 + a_2\,(u - u^*)^2$$

The preferences of the authorities



- Indifference curves are concave
- Slope expresses relative importance attached to fighting inflation versus fighting unemployment

Phillips curve and the natural rate

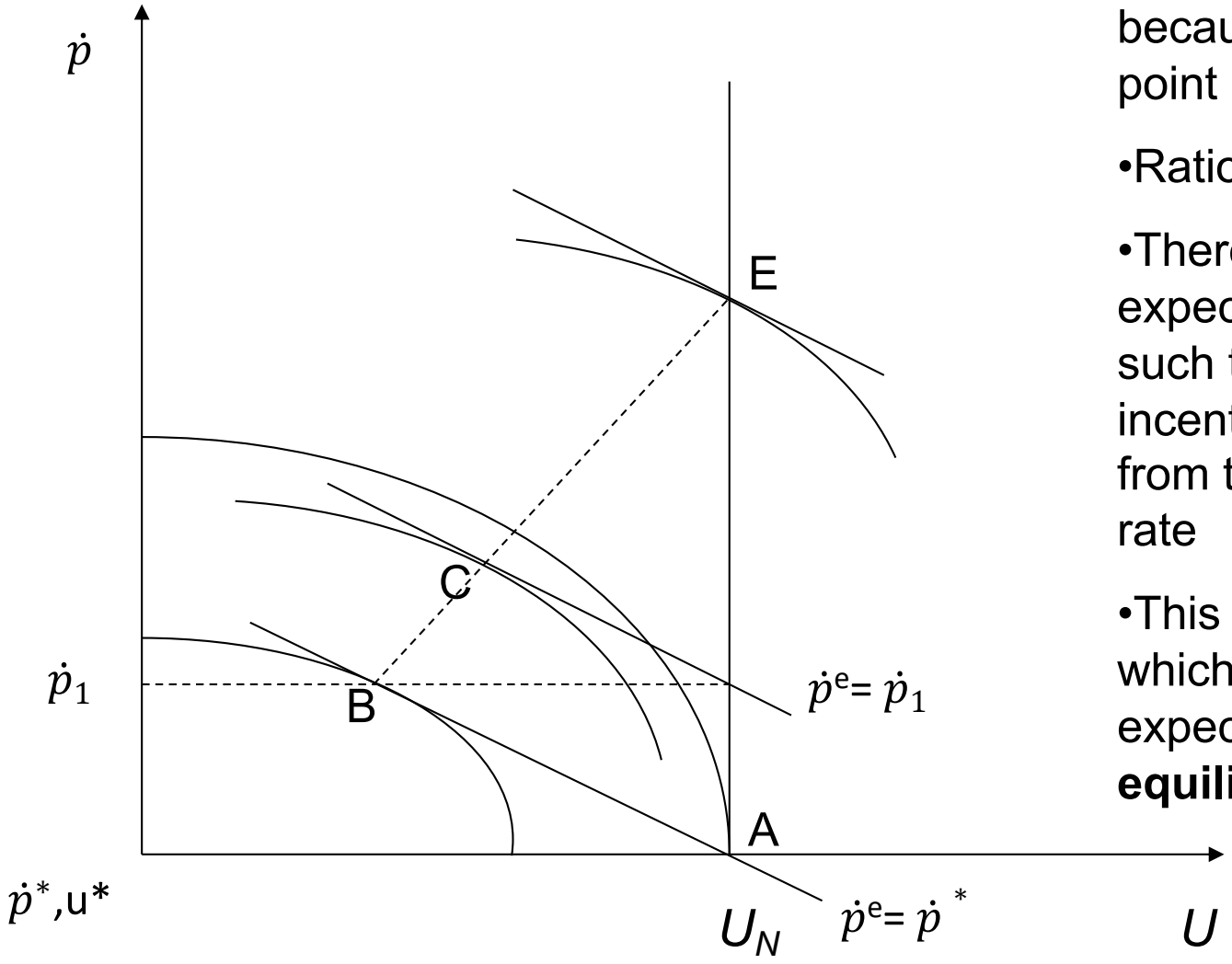


Short run: $\dot{p} = c - bu + \dot{p}^e$

Long run: $u = \frac{c}{b} = u_N$

- There is a short-term trade-off between inflation and unemployment for every level of expected inflation
- The vertical line represents the 'long-term' vertical Phillips curve. It is the collection of all points for which $u=u_N$
- This vertical line defines the natural rate of unemployment U_N

The equilibrium inflation rate



- Announcing a p* inflation policy (point A) is not credible because authorities prefer point B to A
- Rational agents know this
- Therefore they will set their expectations about inflation such that authorities have no incentive any more to deviate from the announced inflation rate
- This is achieved in point E, which is the rational expectations **time consistent equilibrium**

Model

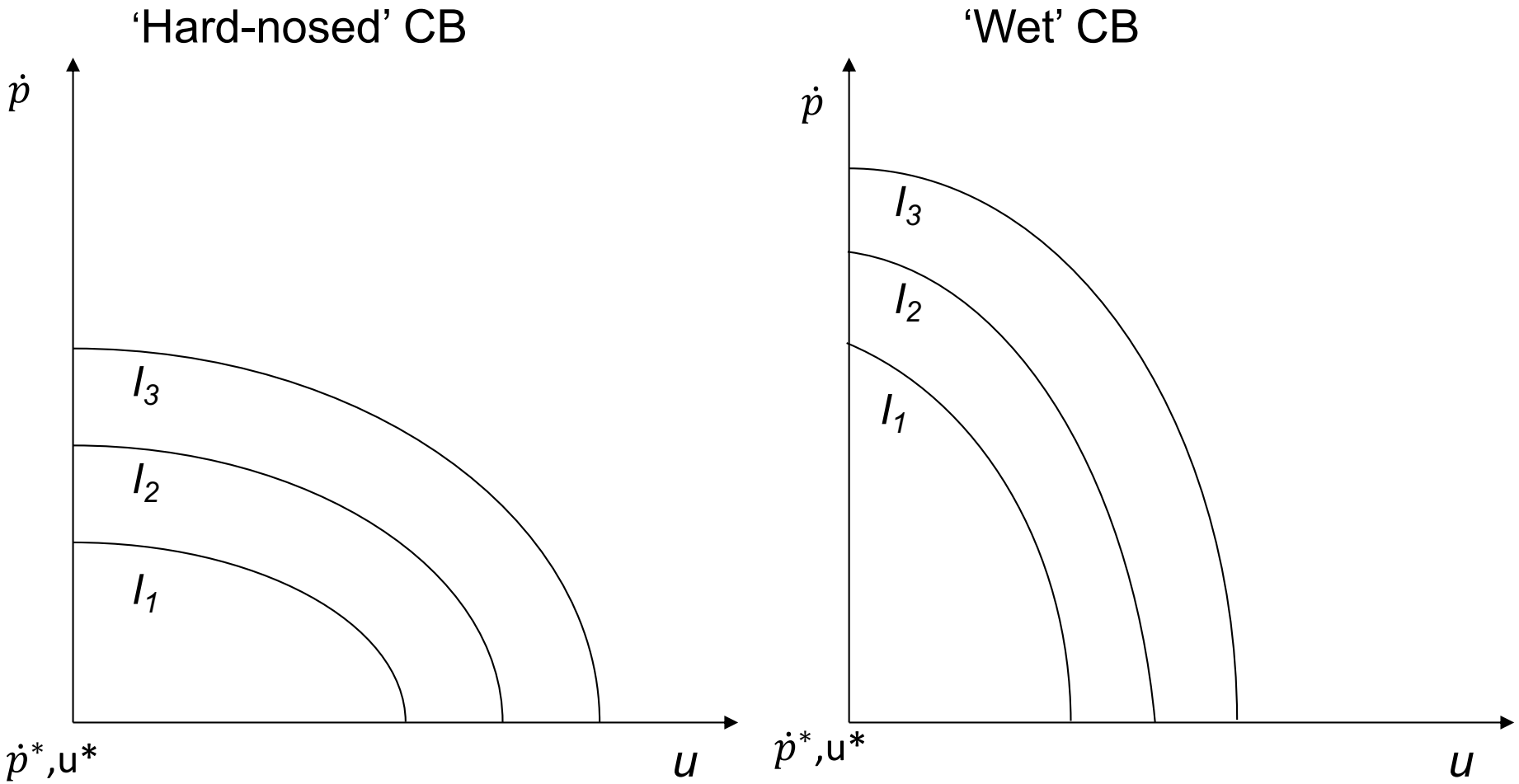
- See pdf

How is the discretional equilibrium achieved?

- The role of expectations:
  - naïve (B)
  - rational (E)
  - adaptive (A,B,C,E)

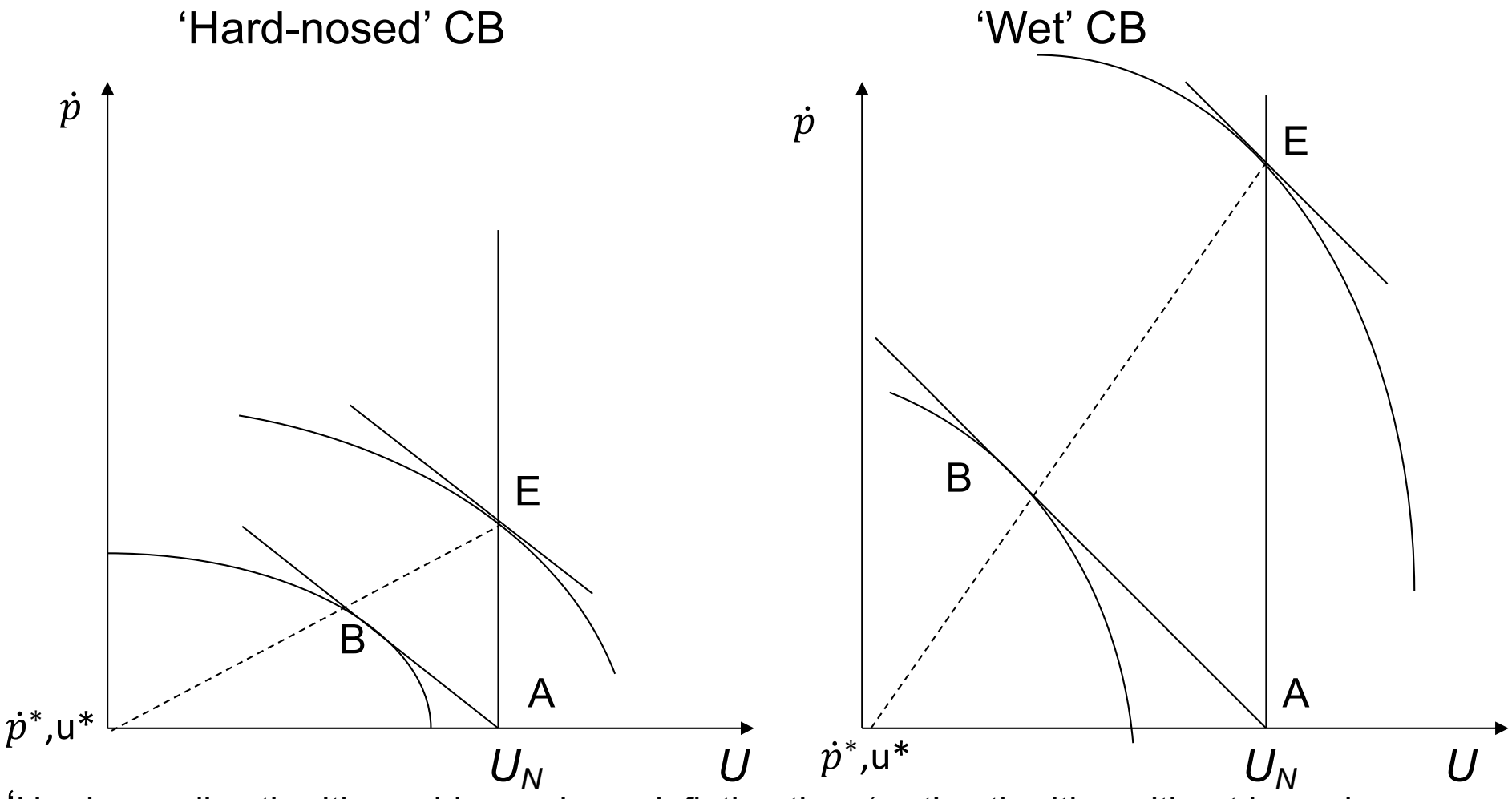
- Note: the equilibrium E is sub-optimal with respect to A: in equilibrium, there is an **inflation bias**
- Recall: the related debate, on «rules vs discretion» has to do with this result and asks if authorities were then to follow rules rather use monetary policy discretionally.
- Note: this model is based on the assumption of rational agents with strategic behaviour.

The preferences of the authorities



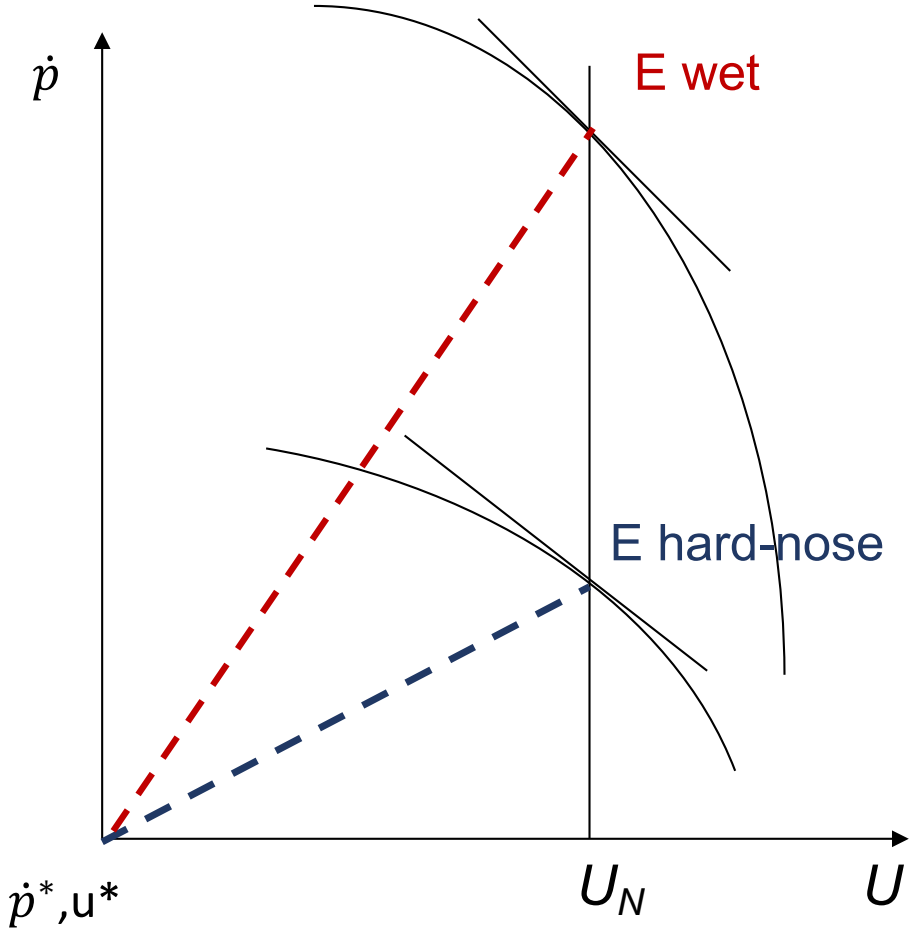
- ‘Hard-nosed’ government attaches a lot of weight to fighting inflation
- ‘Wet’ government attaches a lot of weight to fighting unemployment

The equilibrium differs with authorities' preferences



'Hard-nosed' authorities achieve a lower inflation than 'wet' authorities without imposing more unemployment in the long run

The equilibrium differs with authorities' preferences



- The dashed lines are the loci of points of the optimal policy of the authorities obtained for all possible values of the constraint.
- They are also called **reaction curves.**
- The slope of the reaction curve depends on the authorities' preferences

Conclusion to BG model

Discretionary policymaking can make everyone worse off compared with a situation in which the government could commit to a particular policy path.

Note: Kydland-Prescott work was not related only to monetary policy but to all types of government policy.

The 1983 works of Barro and Gordon focused on the issue of monetary policy and highlighted the role for monetary rules as a potential means to overcome the time inconsistency problem in monetary policy.

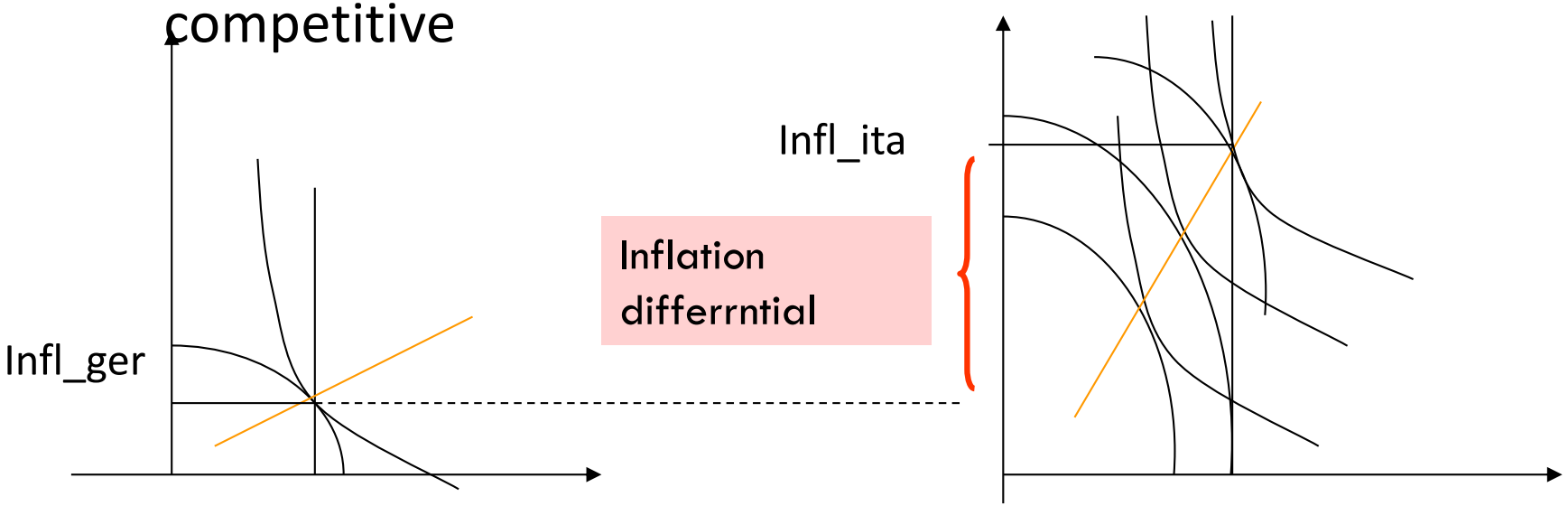
Later suggestions to solve time-inconsistency in terms of delegation (Rogoff 1985)

Further notes on monetary authorities preferences

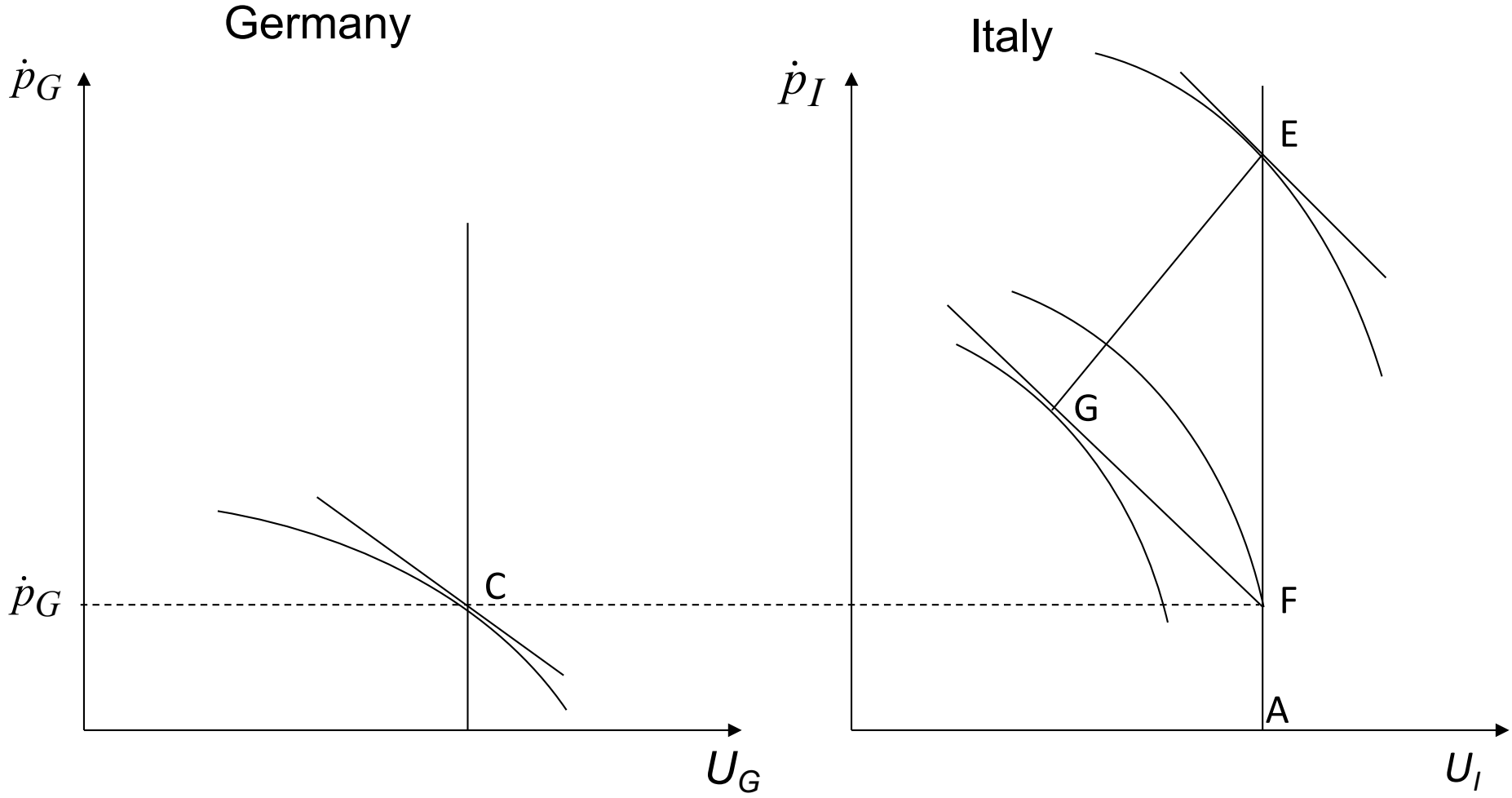
- Super conservative
  - b=0 in the loss function
- Target conservative
  - $u^*=u_N$

Barro-Gordon model for open economies

Two countries: one has a hard-nosed CB (e.g. Germany), with a lower inflation equilibrium; the other has a wet CB (e.g. Italy) with equilibrium with higher inflation. An inflation differential is created→ Italy would have to devalue repeatedly to remain competitive



How can Italian authorties make point F a credible announcement?



Note: in the absence of this credibility, inflation in Italy remains higher and therefore recurring devaluations are needed to maintain price competitiveness

How can Italian authorities make point F a credible announcement?

- Switch to the fixed exchange rate - you can not gain competitiveness with depreciation.
- However, this may not be credible (although more credible than the announcement to keep the inflation rate constant). There remains the temptation for the Italian authorities to increase inflation by surprise. Or to devalue and change the announced parity
- How to make the choice of point F totally credible?

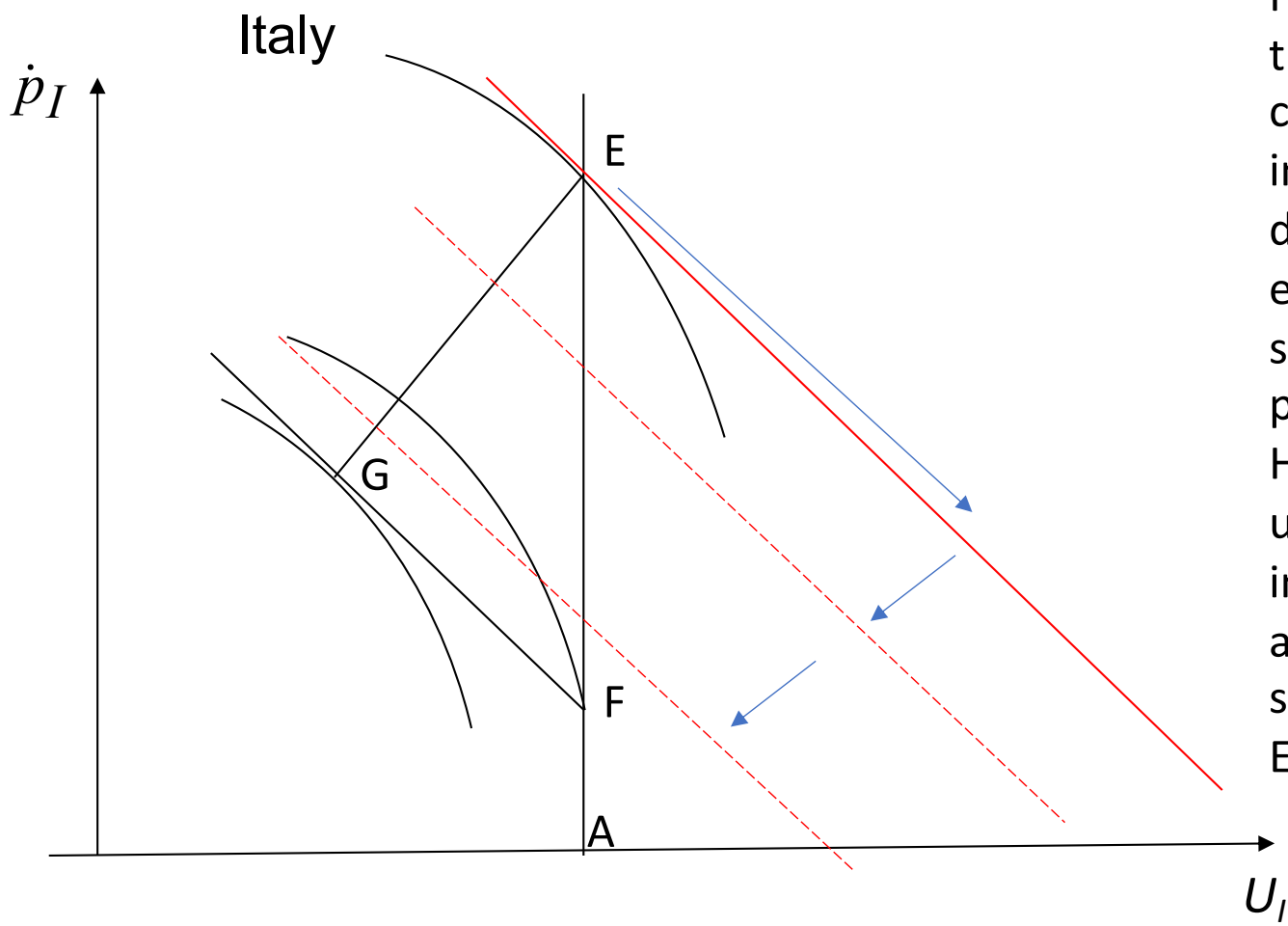
Abandon the lira in favour of the DM?

- Germany's lower inflation could also be enjoyed by Italy.
- Italian authorities could 'borrow' Bundesbank preferences and credibility
- Analogy with the 'dollarization' of the small economies of Latin America or some Eastern European countries with the euro (Kosovo, Montenegro).
- The choice to create a Monetary Union between GER and ITA is more complex because a MU will have a NEW Central Bank that will have to create its own reputation (see below)
- The Italian authorities 'tie their hands' (Giavazzi and Pagano) with regard to monetary policy
-

How was the actual movement from E to F achieved?

- The movement from point E to point F was not painless nor fast
- One of the Maastricht convergence criteria is in fact about reducing inflation to max 1.5 pp higher than the average inflation of the three countries with the lowest inflation
- Countries had almost 10 years to converge to that level
- The policies adopted by the countries that aimed to join the MU were also a signal that policy makers' preferences were changing

How was the actual movement from E to F achieved?



From E down along the short-run Phillips curve and only as inflation expectation declined the economy started shifting towards point F.
Hence the rate of unemployment initially increased and became a serious problem in Europe

Some further conclusions

Joining a MU can also bring a gain if the new CB is more credible; this can help reducing inflation

This was the case for some European countries

However, in a MU the independence of the CB from governments, if the MU lacks a budgetary union or other forms of protection, can have negative repercussions as it may allow self-fulfilling debt crises.(Next topic)